EART11200: Natural Scientist's Toolkit





This week we will cover the Richter scale, and some atmospheric properties. First, consider logarithms.

Atmospheric Science 2

The pressure reduces with altitude according to an exponential law, known as the hydrostatic equation

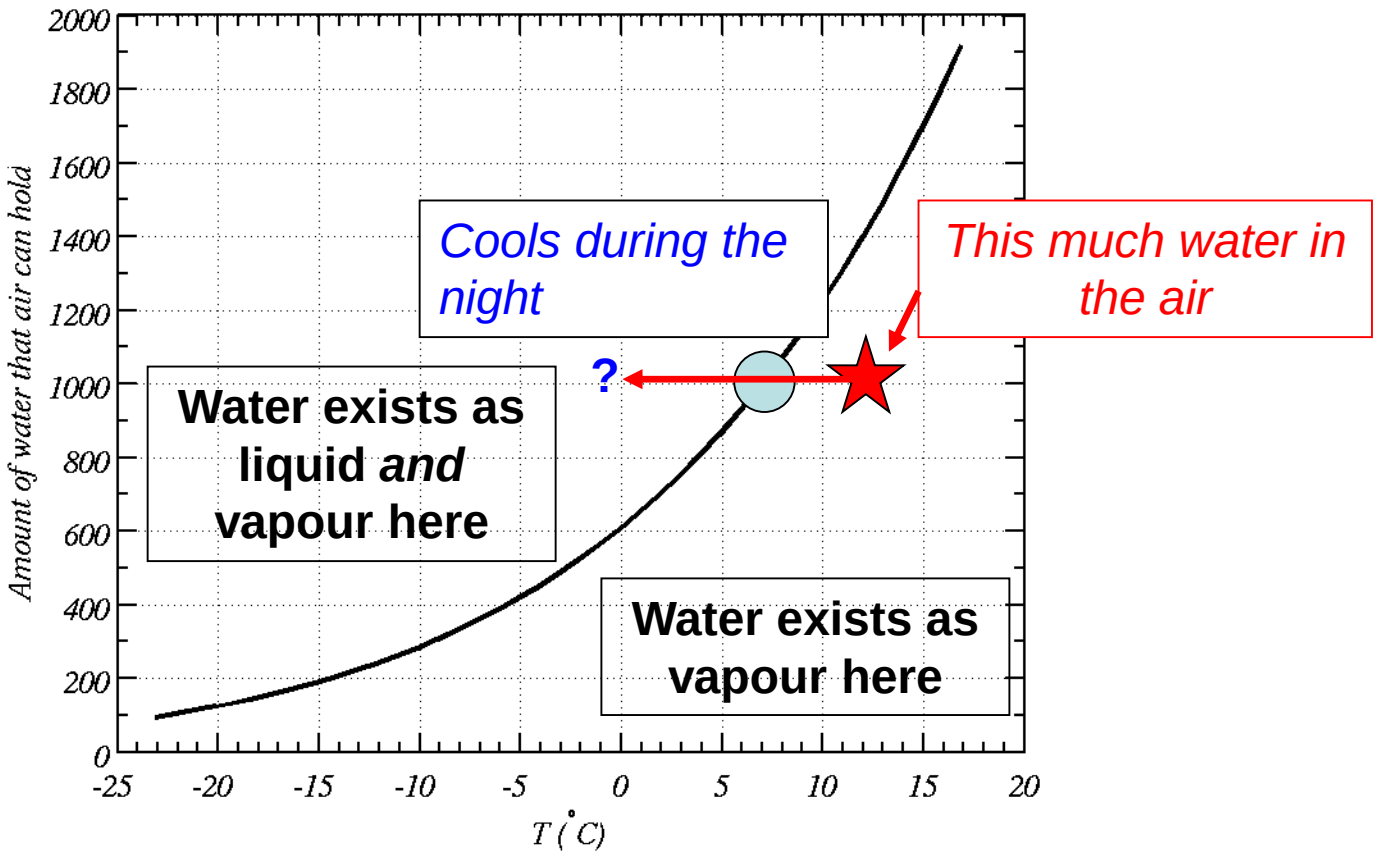
Atmospheric Science 2: In the atmosphere the pressure varies with altitude approximately according to the hydrostatic equation.

$$P = P_0 \exp\left(-\frac{z}{H}\right)$$

where P is the pressure at height z, P_0 is the initial pressure, z is the height in metres and $H = 8000$ m is the 'scale height' of the atmosphere.

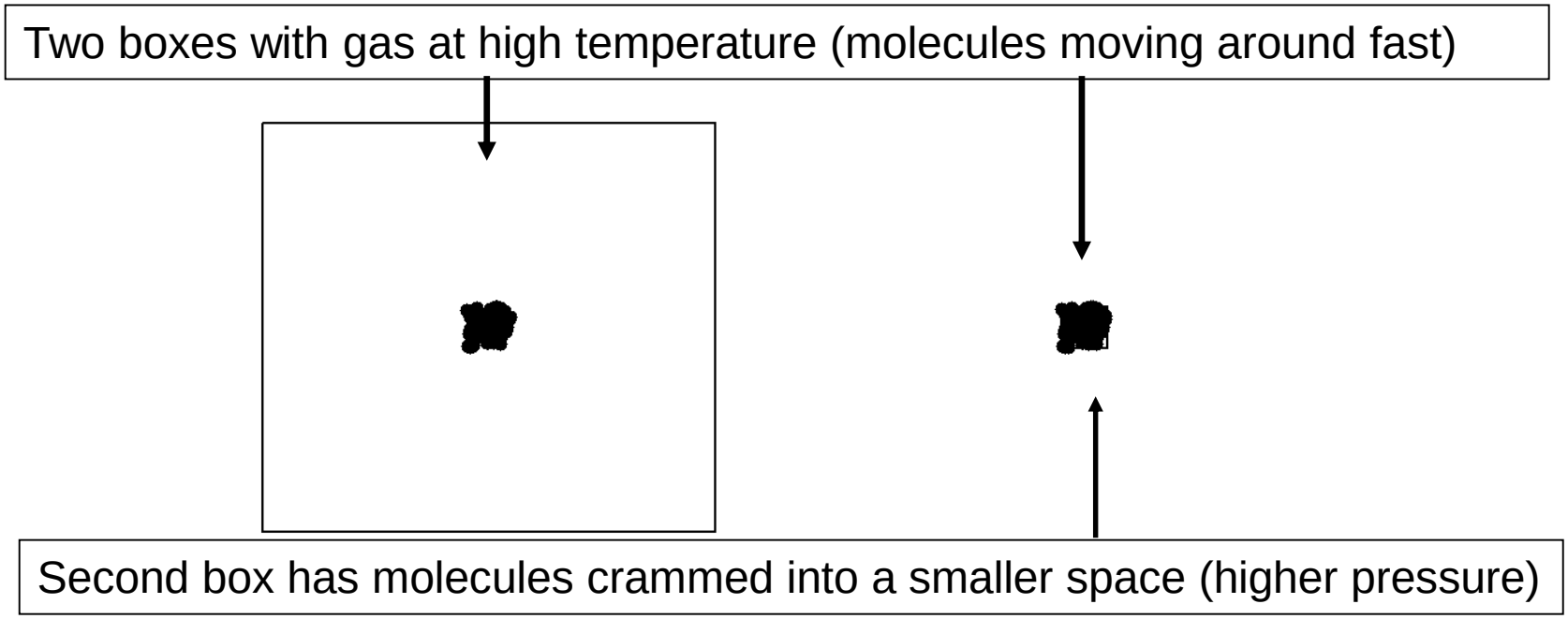
But for rising parcels of air, temperature and pressure are related by a conservation law…

Warm air holds more water vapour than cold air

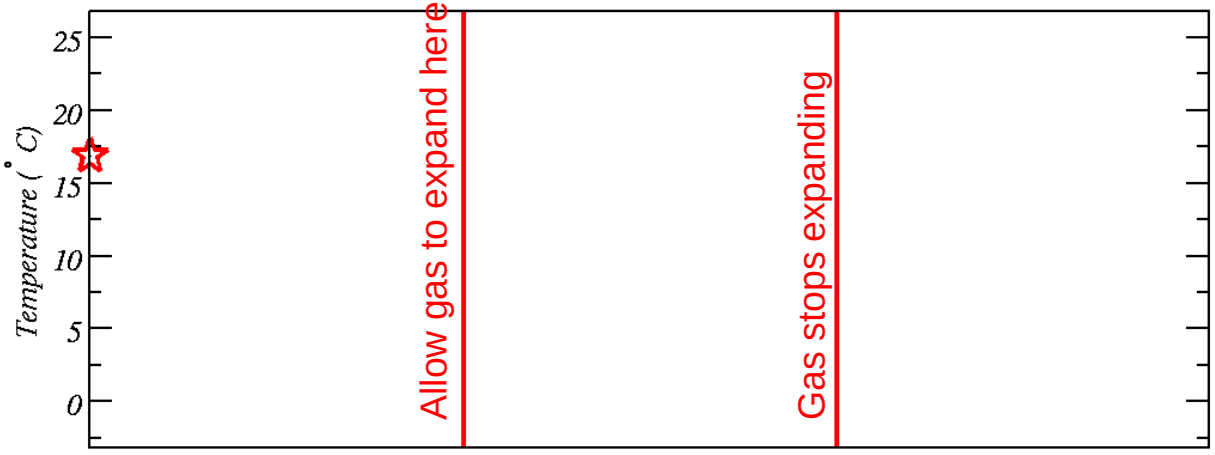


Gas `pushes' against air around it and gas molecules slow down

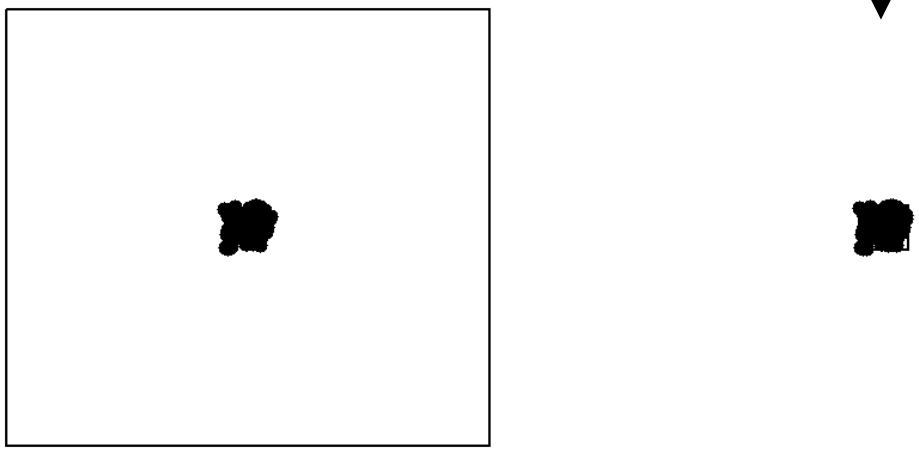
1. Using **Kinetic Theory** how does the speed (Kinetic Energy) of molecules affect their temperature?
2. Do you know a law describing how energy and work are related?



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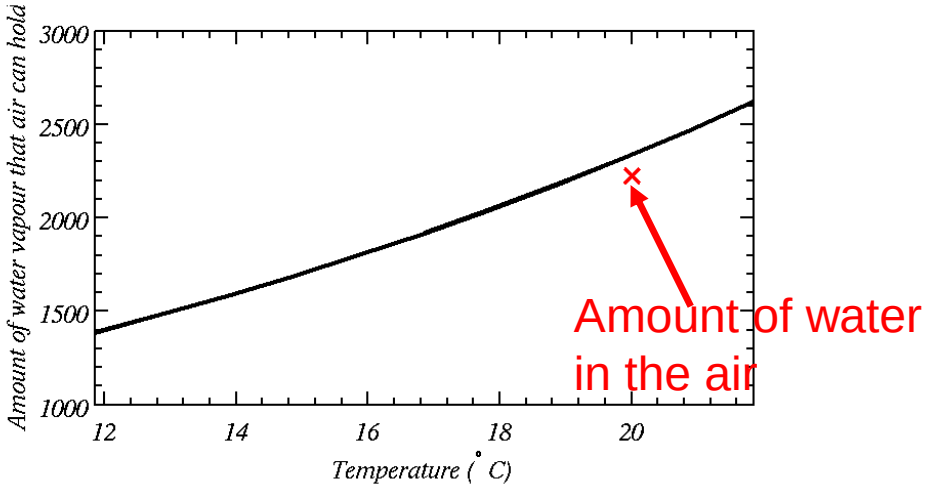
We are going to allow the gas molecules **in second box** to `push' against the surrounding air and slow down as a result (temperature is reduced)



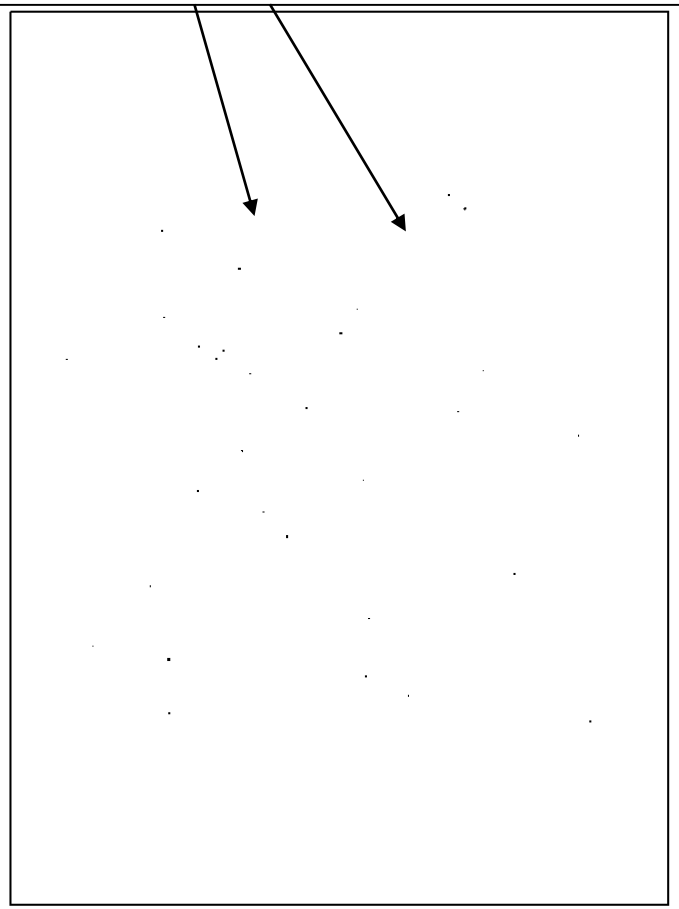
Cloud Condensation Nuclei

As the air cools during the expansion, water vapour eventually condenses onto ***Cloud Condensation Nuclei*** (a type of small particle suspended in air) and forms a cloud

Some small particles do not act as ***Cloud Condensation Nuclei***



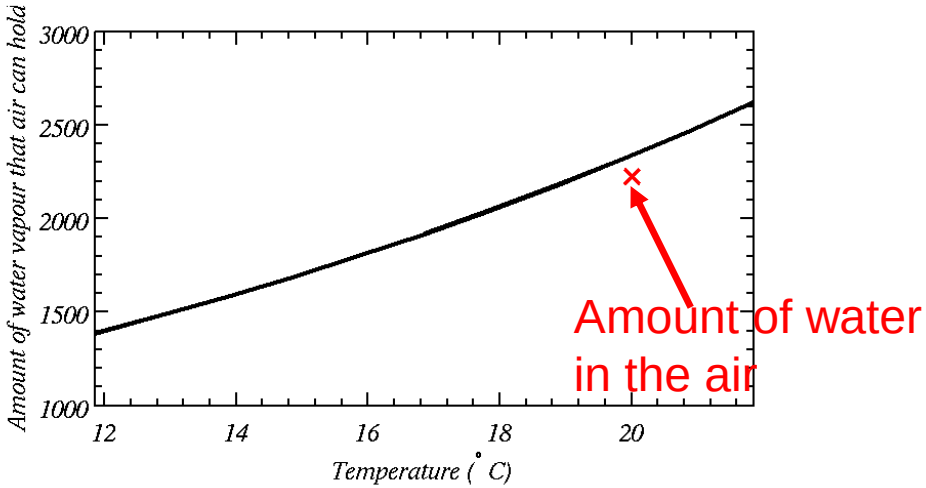
Box with cloud condensation nuclei in it



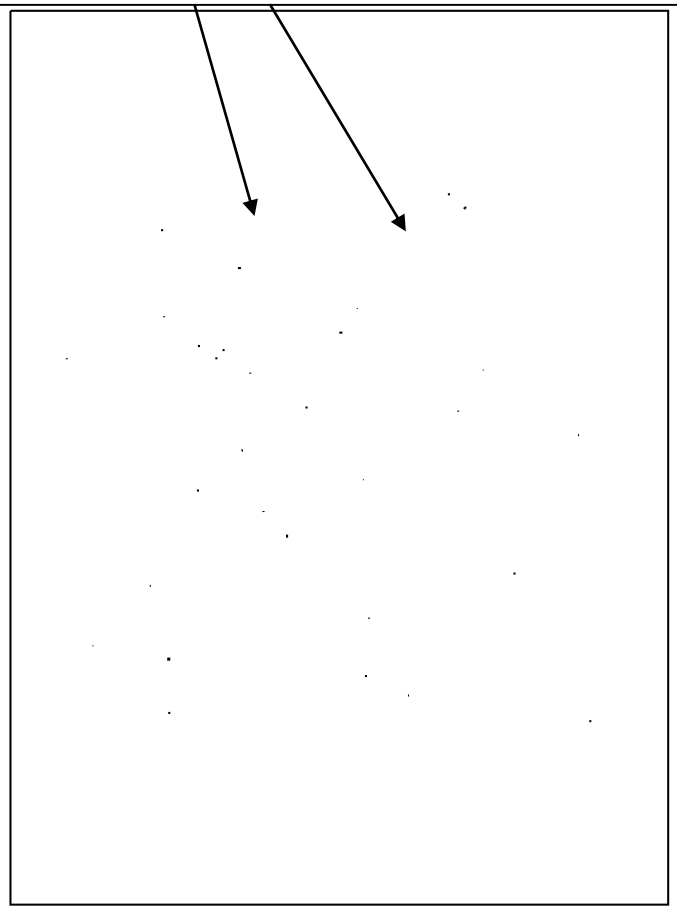
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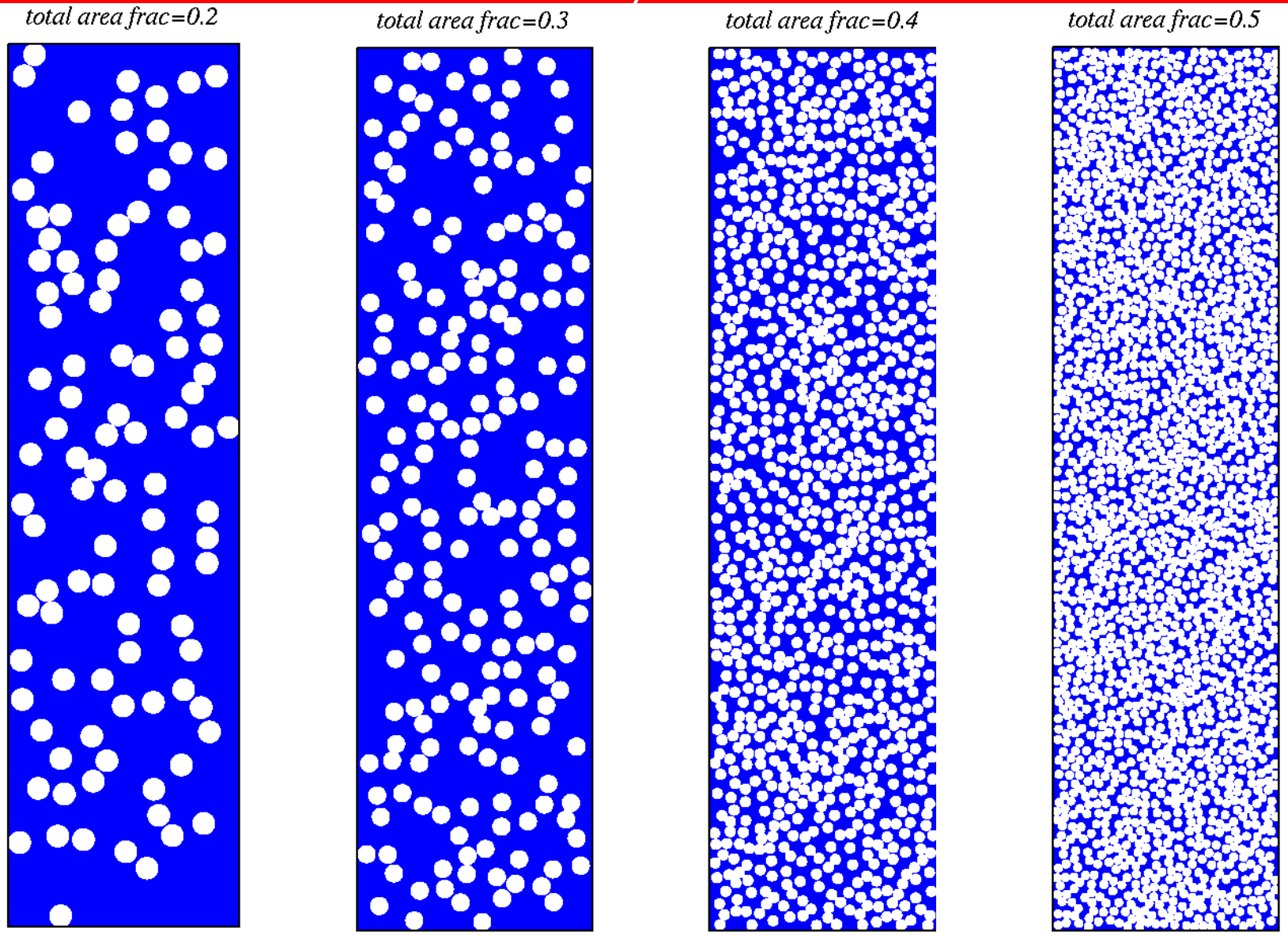


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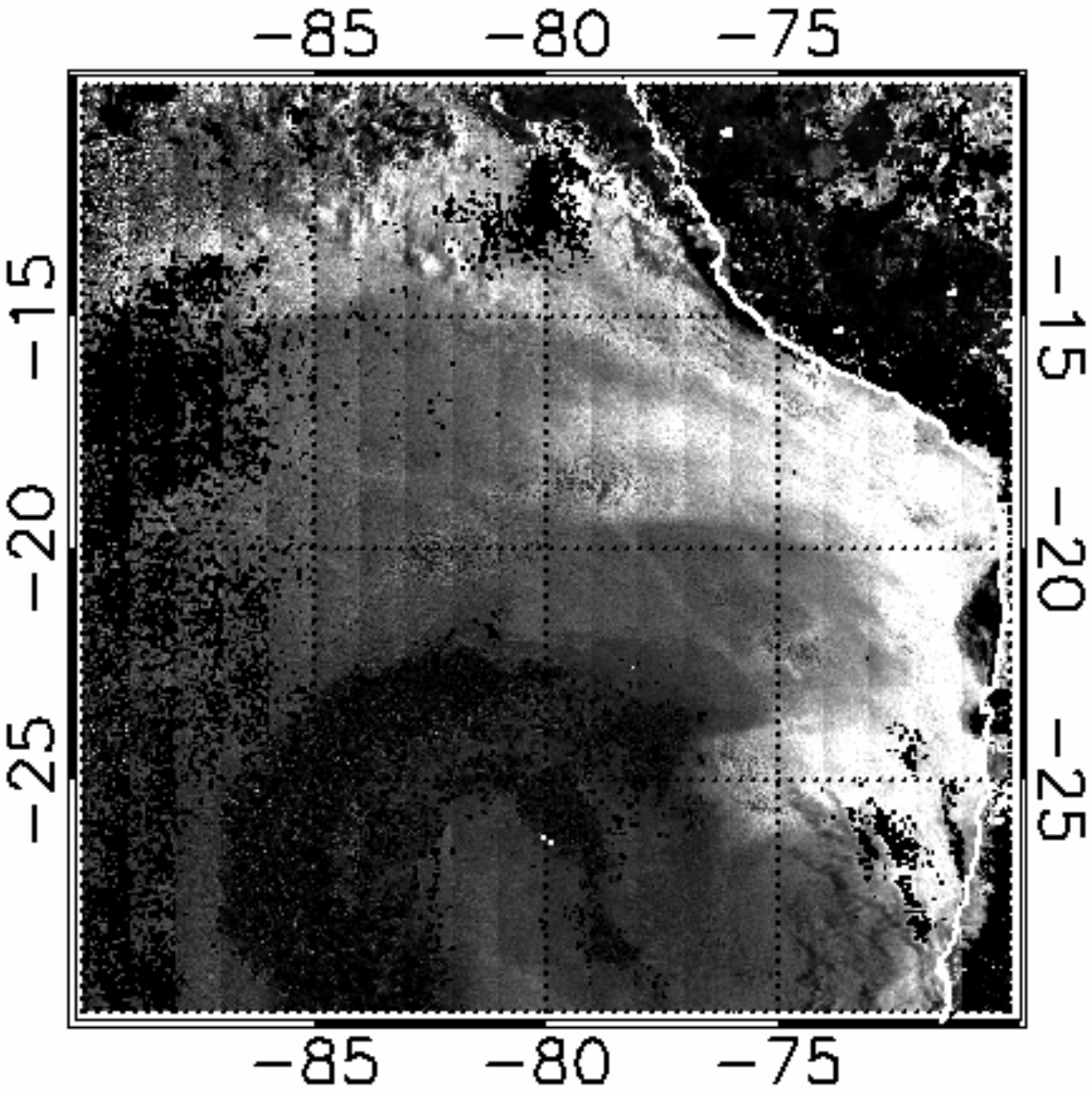


The number of cloud condensation nuclei is important to how much sun-light is reflected

what do you notice?



What do you think the `straight' features are?



c/o Grant Allen

Atmospheric Science 2

By considering conservation laws, atmospheric scientists have shown that…

Additionally, under certain conditions, the quantity θ, given below, does not change

$$\theta = T\left(\frac{10^5}{P}\right)^{0.286}$$

where T is the temperature and P is the pressure.

So if you calculate the initial value of θ it will stay the same even

when pressure changes

In the problem we ask the question “what pressure will it be if the temperature reduces by 5 K?”

In the practical / this week

- Do supplementary material (logarithms and exponentials)
- Do main problems: and 1 & 2 (don’t need to do 3)
- Homework: Blackboard exercises
  - Exponentials and Logarithms